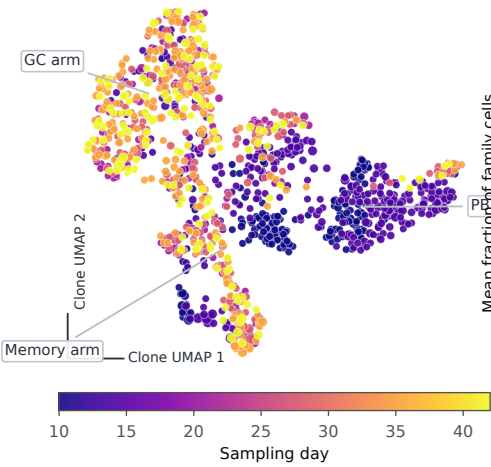


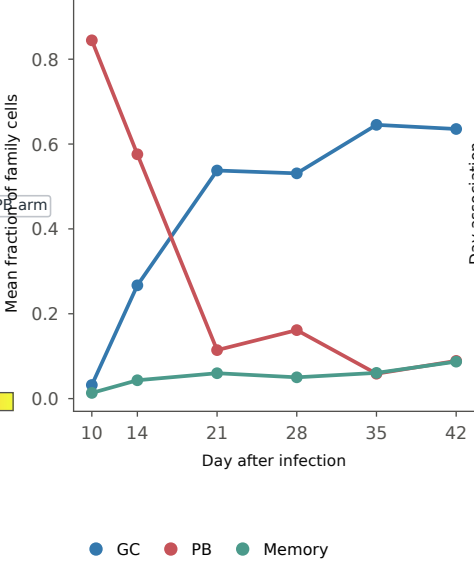
Supplementary Figure 12 | The later-infection clone map encodes the infection time course

Terminal samples; day is a donor-level label (unstratified null); LZ score is expression-derived (descriptive). Clone map: UMAP k=50, min_dist 0.5 (see D).

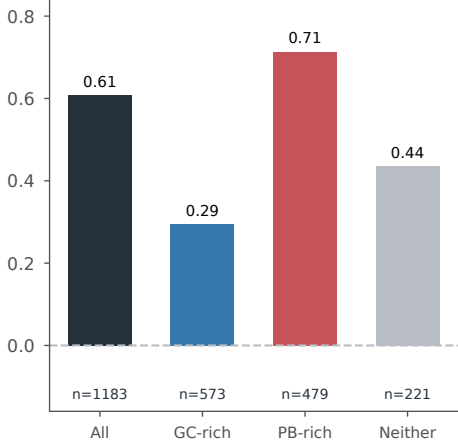
A Later-infection clone map coloured by sampling day



B Reliable-clone composition across the time course



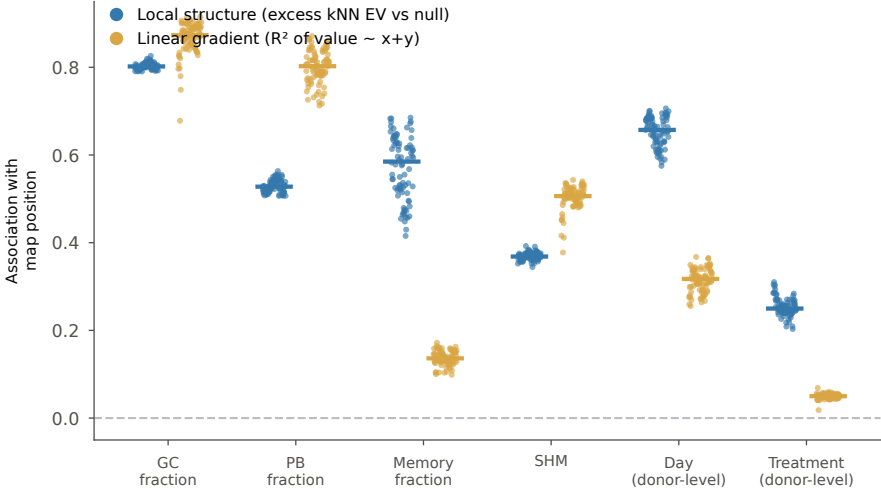
C Day structure persists within state strata



All strata p=0.003 (300 permutations). Donor-level label; unstratified null; state strata defined by family fractions.

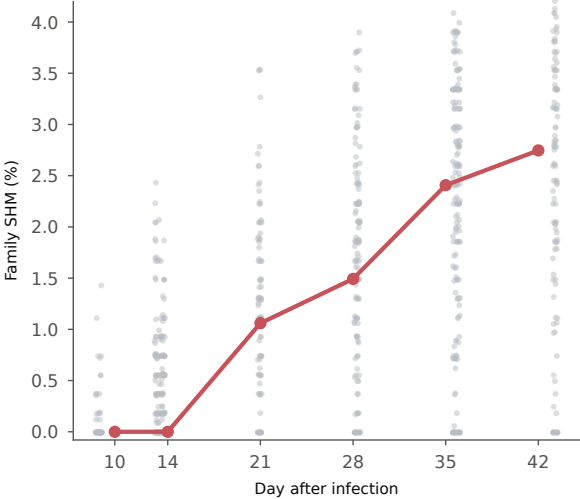
Means over reliable clones sampled per day, including controls. Terminal samples; per-mouse infected summaries in S13.

D Map associations across 85 embedding parameter sets



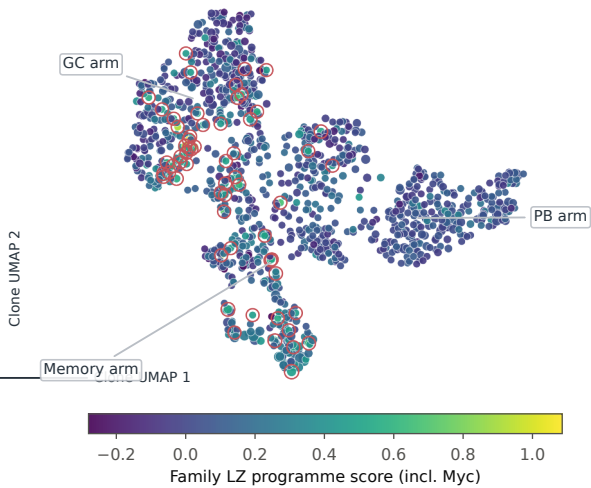
85 UMAP parameter sets (n_neighbors/min_dist/spread/seed), map rebuilt from committed family features each time; bars = median.

E V-region mutation burden across sampled infection dates



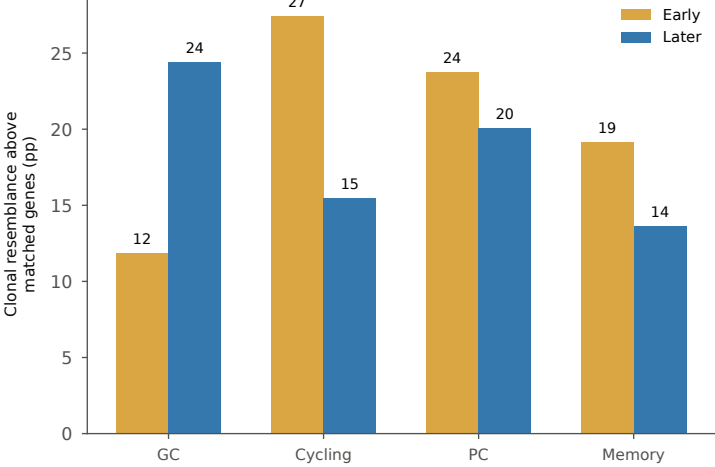
Red: median per day. Median family SHM rises 0.07% (d10) to 2.50% (d42); map gradient linear R² = 0.50 (panel D, SHM gold).

F LZ-programme families sit at the base of the GC arm



Expression-derived, descriptive. Red rings: top 5% LZ families.

G GC and output programmes are clonally coherent above background



Family-level expression resemblance of each programme versus expression-matched background genes (both infection cohorts).